The chiral projective classification from three common interfaces

The human_compression project

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Abstract

We recover the full principal classification and constructions of Leemans–Moerenhout–O'Reilly-Regueiro, in the exact arXiv-v1 edition. The common theory supplies anchored rank descent, two affine ends and one exceptional trace quadratic. These give all the source's prime-field branches and pointed class counts, with the omitted residue restored. The project's earlier accepted primitive-square and translated-subfield proofs also reproduce the subsequent completion for odd composite degrees. The whole shared paper, complete finite premises and retained classical inputs are explicit dependencies; no partial branch is presented as the whole source.

Write $L(q) = \mathrm{PSL}_2(q)$ and $P(q) = \mathrm{PGL}_2(q)$, $q = p^d \geq 4$. For an odd prime $p \neq 5$ with 5 a square, put

$$D_4 = \{1 + \sqrt{5}, 1 - \sqrt{5}\}, \quad D_{55} = \{(7 + 5\sqrt{5})/2, (7 - 5\sqrt{5})/2\}, \quad D_{35} = \{3 + 2\sqrt{5}, 3 - 2\sqrt{5}\}.$$

For such a list D, let $s(D)$ count its nonzero square entries in F_p. If 5 is not a square set all three counts to zero; also set $s(D_4) = 0$ unless 2 is a square.

Theorem 1. *Neither $L(q)$ nor $P(q)$ is the full group of a chiral polytope of rank three or at least five. In rank four:*

1. *$P(q)$ occurs exactly for $q > 4$.*
2. *For even q, the same holds for $L(q) = P(q)$. For $q \equiv 1 \pmod 4$, $L(q)$ occurs exactly for $q > 9$.*
3. *If $q = p^d \equiv 3 \pmod 4$ with $d > 1$ a prime power, $L(q)$ does not occur.*
4. *If $q = p \equiv 3 \pmod 4$ is prime, every type, up to duality, is $(5, 3, 4)$, $(5, 3, 5)$ or $(3, 5, 3)$, and their numbers of ordered isomorphism classes are respectively $2s(D_4)$, $2s(D_{55})$, $2s(D_{35})$. In particular existence is equivalent to at least one count being positive.*
5. *If $q = p^d \equiv 3 \pmod 4$ and $d > 1$ is not a prime power, $L(q)$ occurs. This final clause reproduces the project's previously accepted completion, not a result asserted as proved in the source edition.*

Proof. The simultaneous-reversal theorem of [U] excludes rank three; its anchored involution theorem and subfield envelope exclude all ranks at least five. The pointed construction and the complete A_5, A_6 exclusions give the first two rank-four clauses. The nonsingular trace extension formula puts a marking over a quadratic extension of the compositum of the two proper end fields; when $d > 1$ is an odd prime power this compositum is proper and the quadratic extension is absent. This is the common odd-degree exclusion.

For prime $p \equiv 3 \pmod 4$, neither end fixes a projective point. The common subgroup reduction and full exceptional local-pair census leave exactly the three displayed types. The trace-extension and collision theorems then give all the stated counts, generation, intersection conditions and chirality. In particular $p = 11$ contributes no $(3, 5, 3)$ class, while $p = 19$ contributes two $(5, 3, 5)$ classes. These

small conclusions are exact finite premises, not assumptions borrowed from the source's table. Finally separate the prime-power factors of a non-prime-power odd d into coprime $e_1, e_2 > 1$ and apply the common coprime-subfield construction. Every part has both its necessity and sufficiency supplied in the paid common paper. □

For direct comparison with source Theorem 1(4), the five existence conditions in its prime branch are

(a)	$p \equiv 11, 19$	$\pmod{20}$,	$p \bmod 11 \in \{1, 3, 4, 5, 9\}$, D_{35} both squares;
(b)	$p \equiv 11, 19$	$\pmod{20}$,	$p \bmod 11 \in \{2, 6, 7, 8, 10\}$;
(c)	$p \equiv 11, 19$	$\pmod{20}$,	$p \bmod 19 \in \{1, 4, 5, 6, 7, 9, 11, 16, 17\}$, D_{55} both squares;
(d)	$p \equiv 11, 19$	$\pmod{20}$,	$p \bmod 19 \in \{2, 3, 8, 10, 12, 13, 14, 15, 18\}$;
(e)	$p \equiv 31, 39$	$\pmod{40}$.	

The restored residue 10 in (b) is present in the source's Theorem 11 but missing from its introductory Theorem 1. The common discriminant products $-11, -19, -4$ and quadratic reciprocity prove precisely these conditions. The case $p = 19$ is already included by (b). As in the source, both enantiomorphic ordered classes are counted when distinct; the displayed $(5, 3, 4)$ counts also hold for its dual $(4, 3, 5)$. These are the source proof's ordered generating-tuple classes under $\mathrm{Aut}(G)$; no further identification by changing the base flag to the opposite flag orbit is imposed, and no separate unbased-poset census is inferred from them.

1 All pointed ends, not only one existence example

The following includes the full counts of source Theorems 4 and 5. For a positive integer $k > 2$ require

$$k \mid (q - 1), \qquad k \nmid p^e - 1, \quad k \nmid p^e + 1 \quad (0 < e < d).$$

For odd q also require k even. If the group is $L(q)$ require $k \mid (q - 1)/2$; if it is $P(q) \neq L(q)$ require $k \nmid (q - 1)/2$. In characteristic two these two conditions are replaced by equality $P(q) = L(q)$, with no even-k condition. There are exactly $\varphi(k)/d$ ordered chiral markings whose two ends each fix a projective point. Their common type is (k, k, k) in characteristic two or when $4 \mid k$, and $(k/2, k, k/2)$ when $k \equiv 2 \pmod 4$.

Here is the complete source marking, with an element t of order k:

$$a = \begin{pmatrix} -t^{-1} & 1 \\ 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 & 0 \\ 1 + t & -t \end{pmatrix}.$$

The common pointed theorem proves both ends are $F_q^+ \rtimes \langle -1, t \rangle$, their intersection is the common diagonal group, and they generate the whole specified projective group. It proves the normal form is exhaustive and the only identifications are Frobenius on t. Thus the construction gives all classes and every claimed count. The condition excluding a Frobenius inversion of t excludes a regular marking, not necessarily a proper generated subgroup. For example $q = 25$, $|t| = 6$ gives all of $L(25)$, with end orders 150 and intersection six, but its Frobenius reversal makes it regular. This corrects that proof route in the source without changing its chiral list.

2 Every exceptional prime-field class is constructive

The counts in Theorem 1 are supplied by actual matrices, rather than only a square criterion. Let a, b, c now denote chosen determinant-one traces, and put $d_0 = a^2 + b^2 - 4$. For type $(5, 3, 4)$ choose $a = (-1 \pm \sqrt{5})/2$, $b = 1$, $c = \sqrt{2}$; for $(5, 3, 5)$ choose $a = c = (-1 \pm \sqrt{5})/2$, $b = 1$; for $(3, 5, 3)$ choose

$a = c = 1$, $b = (-1 \pm \sqrt{5})/2$. Choose $U^2 + V^2 = d_0$, and for either square root z of the associated nonzero discriminant set

$$u = (U + a)/2, \quad w = (V - b)/2, \quad A = \begin{pmatrix} u & w + b \\ w & a - u \end{pmatrix}, \quad B = A^{-1}J, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

With $R = (-bcV + zU)/d_0$, $S = (bcU + zV)/d_0$, put

$$C = \frac{1}{2}\begin{pmatrix} R + c & S \\ S & c - R \end{pmatrix}.$$

The common trace theorem proves all determinant, trace and involution identities, pair uniqueness and the exact number of roots. Its finite intersection theorem proves that every such nonzero-root extension is polytopal. Prime-field subgroup classification proves generation and the nonzero invariant $2\,\mathrm{tr}(AC) - ac = z$ forbids reversal. For $q \equiv 3 \pmod 4$ these types require $q = p$ prime: the traces and their quadratic extension all descend through an odd-degree ambient field to F_p. This recovers the full scopes of source Theorems 6, 7, 9 and 11 without invoking their borrowed all-field quotient lists.

As a literal witness to the introductory omission, over F_{571} take

$$A = \begin{pmatrix} 0 & 481 \\ 184 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 481 & 1 \\ 0 & 387 \end{pmatrix}, \quad C = \begin{pmatrix} 113 & 411 \\ 411 & 459 \end{pmatrix}.$$

These have type $(3, 5, 3)$; the two ends have order 60, their intersection order five, and the full group has order 93 084 420. The reversal obstruction is $\mathrm{tr}(AC) - 1/2 = 550 \neq 0$. Here $\sqrt{5} = 24, 571 \equiv 10 \pmod{11}$ and the other introductory clauses fail: 2, 349 and 229 are nonsquares, the latter two being D_{55}. The uniform proof above implies these conclusions, and the separate complete permutation validation records the exact witness. No new conjecture answer or historical-priority claim follows from this version-specific correction.

3 The earlier odd-composite-degree completion

For odd coprime $e_1, e_2 > 1$, the common primitive-square theorem supplies primitive $j_i \in F_{p^{e_i}}$ and a nonzero square root $D^2 = u^2v^2 - 4(u^2 + v^2)$, $u = j_1 + j_1^{-1}$, $v = j_2 + j_2^{-1}$. The common coprime-subfield theorem then gives the exact source matrices

$$\alpha = \frac{1}{2}\begin{pmatrix} u & u + 2 \\ u - 2 & u \end{pmatrix}, \quad \beta = \frac{1}{2}\begin{pmatrix} 2 + u & u \\ -u & 2 - u \end{pmatrix}, \quad \gamma = \frac{1}{2u}\begin{pmatrix} v(u - 2) & D \\ D & v(u + 2) \end{pmatrix}.$$

Their type is $((p^{e_1} - 1)/2, p, (p^{e_2} - 1)/2)$. The scalar in γ restores determinant one to the source's final projective representative. The common proof verifies the shared unipotent intersection before generation, then excludes every proper maximal overgroup and every semilinear reversal. The norm-polynomial proof includes equal and nested degrees, the complete integer reduction and all eleven finite certificates. This material is fully reproduced and paid in [U], with the prior phase267 acceptance recorded; it is not an unpriced citation to an earlier proof bank and earns no new question credit.

4 Exact source and evidence scope

The source is the complete twenty-five-page arXiv:1606.08017v1 edition, 26 June 2016. No separate collation of its later journal edition is claimed. We recover all its own principal classification and construction clauses, with the exact corrected residue and characteristic qualifications. The shared collision program also repairs its Lemma 5.12 calculation: four of the twelve claimed trivial A_4-collision quotients are A_5, but its intersection conclusion outside $L(4), L(5), L(19)$ is unchanged. The source's

general abelianization wording means a quotient, not a subgroup; its finite cyclic application is unaffected. The signed-order formula in Lemma 4.1 uses odd characteristic; in characteristic two $-t = t$. The rank-three small identifications are the standard $P(3) = S_4$, $P(2) = S_3$, correcting the printed sentence after Theorem 4.

Theorems 8 and 10 are quoted all-field Coxeter-quotient classifications from Jones–Long and Jones–Long–Mednykh. They are not invoked in our principal proof and their entire auxiliary all-field statements are not newly rederived. Tables 1 and 2 are motivating finite census data, including the 44 objects at $q = 169$; their full atlas computation is not newly regenerated. The source map distinguishes these ancillary items from every recovered principal clause and every explicit construction used above. Its diagrams illustrate a replaced proof route. No claim of literal all-assertions equivalence is made.

All three previous Topic E applications remain present, with their whole common proof and complete required program. The additional required finite premises and full polynomial JSON data are disclosed, with validation and operational files separate. The nine new Lean algebra statements and reused scoped certificates do not formalize classical classification, character sums, finite GAP quotients or the polytope correspondence. External proof, catalogue and graphic costs remain unknown.

References

[U] The human_compression project, *Common theory for projective markings and rotations*, current complete common paper.

[1] D. Leemans, J. Moerenhout and E. O'Reilly-Regueiro, *Projective linear groups as automorphism groups of chiral polytopes*, arXiv:1606.08017v1, 26 June 2016.

[2] E. Schulte and A. I. Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS 4 (1991), 435–516.